BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ECE 4TH SEMESTER - ALL SUBJECTS PYQS

Exam Date: 29 Oct 2026 - 07 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2024	Analog Communication	104402	B.Tech 4th Semester Examination, 2024
2023	Analog Communication	104402	B.Tech 4th Semester Examination, 2023
2024	Analog Circuits	104401	B.Tech 4th Semester Examination, 2024
2023	ECE Analog Circuit	104401	End Semester Examination - 2023 Semester-IV
2022	Analog Electronic Circuits	110401	B.Tech 4th Semester Exam., 2022
2020	Linear Integrated Circuits and Applications	104504	B.Tech 5th Semester Exam., 2020 (New Course)
2023	Digital Circuits	104403	End Semester Examination - 2023
2024	Digital Electronics	100403	B.Tech 4th Semester Examination, 2024
2023 (Special)	Digital Electronics	100403	Special Examination - 2023

2023 (Regular)	Digital Electronics	100403	End Semester Examination - 2023
2024	Electromagnetic Theory	104404	B.Tech 4th Semester Examination, 2024
2023	Electromagnetic Theory	104404	End Semester Examination - 2023
2020	Electromagnetic Field Theory	100308	B.Tech 3rd Semester Exam., 2020 (New Course)
2024	Semiconductor Physics and Devices	104405	B.Tech 4th Semester Examination, 2024
2023	Semiconductor Physics and Devices	104405	End Semester Examination - 2023 (Semester IV)
2024	Electronics Instruments and Measurement	104604	B.Tech 6th Semester Examination, 2024
2024	Measurements and Instrumentation	110601	B.Tech 6th Semester Examination, 2024
2022	INSTRUMENTATION AND CONTROL	102404	B.Tech 4th Semester Exam., 2022

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

ANALOG COMMUNICATION

Time: 03 Hours | Code: 104402 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):- **[14]**

1. If the carrier of a 100% modulated A.M. wave is suppressed, the power saving will be in percent-
   i. 50
   ii. 67
   iii. 80
   iv. 100

2. Modulation is the process of-
   i. combining audio and radio frequency waves
   ii. reducing distortion in RF amplifiers
   iii. generating constant-frequency radio waves
   iv. improving thermal stability of a system

3. The FM broadcast band is from-
   i. 88 to 108 MHz
   ii. 88 to 108 kHz
   iii. 1600 kHz to 59.75 MHz
   iv. 88 to 108 GHz

4. The standard intermediate frequency for FM is-
   i. 10.7 MHz
   ii. 10.7 kHz
   iii. 4.5 MHz
   iv. 4.5 kHz

5. With 100% modulation. ratio of sideband power to total power transmitted in an amplitude-modulated wave is-
   i. 2/3
   ii. 4/3
   iii. 1/2
   iv. 1/3

6. In an FM signal. as the modulation index increases. the power-
   i. increases
   ii. remains constant
   iii. decreases
   iv. None of above

7. The simplex system is a
   i. bidirectional communication system
   ii. multidirectional communication
   iii. unidirectional communication
   iv. None of the above

8. In _____ modulation the phase of the carrier signal is modulated by the message signal.
   i. pulse
   ii. amplitude
   iii. phase
   iv. frequency

9. In baseband signal the audio frequency signal range is around
   i. 10-20 kHz
   ii. 5-10 kHz
   iii. 0-20 kHz
   iv. 0-20 MHz

10. In super heterodyne receiver. at high frequency the
    i. signal processing should be done
    ii. signal transmission should be done
    iii. bandwidth is low
    iv. Both (i) and (ii)

Q.2 Solve both questions : **[14]**

1. Draw the block diagram of communication system. Explain the concept of frequency mixing.
2. Explain the function of each block of communication system.

Q.3 Solve both questions : **[14]**

1. Explain generation of DSB-SC signal with the help of balanced modulator using diodes.
2. Draw the frequency spectrum of DSB-SC modulation with necessary mathematical expressions. Write short note on Pre-Emphasis and De-Emphasis circuits.

Q.4 Solve both questions : **[14]**

1. Write short note on Pre-Emphasis and De-Emphasis circuits.
2. Draw block diagram of Super-heterodyne AM receiver and explain function of each block.

Q.5 Solve both questions : **[14]**

1. (i) The maximum peak-to-peak voltage of an AM wave is 12 mV and the minimum voltage is 4mV. Calculate the modulation factor. (ii) An audio signal of 1 KHz is used to modulate a carrier 500 KHZ. Determine side band frequencies and band width required.
2. Describe thermal noise and shot noise.

Q.6 Solve both questions : **[14]**

1. With block diagram explain the generation of PWM signals.
2. What are the differences between PAM, PWM, and PPM

Q.7 Solve both questions : **[14]**

1. The output signal from an AM modulator is: $s(t) = 5\cos(1800\pi t) + 20\cos(2000\pi t) + 5\cos(22000\pi t)$. (a) Determine the modulation index. (b) Determine the ratio of the power in the sidebands to the power in the carrier.
2. Explain (i) Signal to Noise Ratio (ii) Figure of merit.
   Discuss about noise effect in PM and obtain expression for figure of merit.

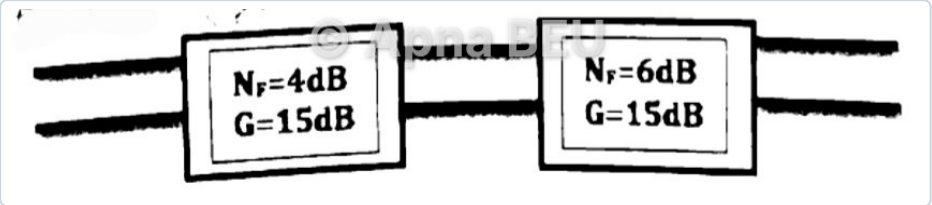
Q.8 Solve both questions : **[14]**

1. Define amplitude modulation and describe it in detail with suitable diagrams.
2. Write the difference forms of Amplitude Modulation, what is the efficiency of the AM?

Q.9 Solve both questions : **[14]**

1. Write short notes our Envelope Detector.

2. Find the overall noise factor for the cascaded noise networks as shown in given figure. [Stage 1: $N_F = 4dB$, $G = 15dB$; Stage 2: $N_F = 6dB$, $G = 15dB$]



BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2023

ANALOG COMMUNICATION

Time: 03 Hours | Code: 104402 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

1. In NBFM, the modulation index is close to
   - i. 1
   - ii. 10
   - iii. greater than 1
   - iv. infinite

2. The maximum power efficiency of an AM modulator is
   - i. 50%
   - ii. 25%
   - iii. 75%
   - iv. 100%

3. Spectral density of white noise is
   - i. Varies with frequency
   - ii. Constant
   - iii. Varies with bandwidth
   - iv. Varies with amplitude

4. If the step size of a system is 0.0721, find its quantization power?
   - i. 3.25×10^{-3}
   - ii. 4.33×10^{-3}
   - iii. 4.33×10^{-4}
   - iv. 3.25×10^{-2}

5. Which of the following frequency is not transmitted in AM transmission?
   - i. Upper side band frequency
   - ii. Carrier frequency
   - iii. Lower side band frequency
   - iv. Audio frequency

6. What is the use of Companding?
   - i. in PCM transmitters to allow amplitude limiting in the receivers
   - ii. in PCM receiver to overcome impulse noise
   - iii. to overcome quantizing noise in PCM
   - iv. to protect small signals in PCM from quantizing distortion

7. Envelope Detector is a/an
   - i. Coherent detector
   - ii. Asynchronous Detector
   - iii. Synchronous Detector
   - iv. Product Demodulator

8. A 1000KHz carrier is modulated by 100Hz and 200Hz waves, then which of the following frequencies cannot be present at the output?
   - i. 1000.1 KHz
   - ii. 999.9 KHz
   - iii. 1000.2 KHz
   - iv. 999.7 KHz

9. For 100% modulation, power in each sideband is ____ of that of carrier.
   - i. 50%
   - ii. 60%
   - iii. 25%
   - iv. 33.33%

10. The frequency of local oscillator
    - i. can be either below or above the RF frequency
    - ii. is below the RF frequency
    - iii. is above the RF frequency
    - iv. is fixed typically at 450KHz

Q.2 Solve this question : **[14]**

1. A radio receiver with equivalent noise bandwidth of 10kHz has a noise figure of 20dB.If input SNR to receiver is 40dB, determine the output SNR. What is equivalent noise temperature if ambient temperature is 27°C.

Q.3 Solve this question : **[14]**

1. Discuss the Armstrong method for the generation of wideband FM.

Q.4 Solve both questions : **[14]**

1. Explain the working of slope detector method used for demodulation.
2. Write notes on PAM, PPM and PWM.

Q.5 Solve both questions : **[14]**

1. Explain need of modulation and its advantage.
2. Derive the expression for power content in an AM wave.

Q.6 Solve this question : **[14]**

1. When a super-heterodyne receiver is tuned to 600kHz,its local oscillator provides the mixer with an input at 1055 kHz. What is the image frequency? The antenna of the receiver is connected to the mixer via a tuned circuit with Q 50. What will be the rejection ratio for the calculated image frequency?

Q.7 Solve this question : **[14]**

1. What is demodulation? Explain square law demodulator.

Q.8 Solve this question : **[14]**

1. Derive the expression of AM modulated wave and draw its waveform.

Q.9 Write short notes on any two of the following: **[14]**

a. Thermal Noise and Shot Noise & Carson Rule.

b. DSB-SC & VSB

c. Difference between Analog & Digital communication

d. Synchronous detection of DSB-SC

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

ANALOG CIRCUITS

Time: 03 Hours | Code: 104401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

1. JFET can be operated with
   - i. negative gate only
   - ii. gate voltage only
   - iii. positive as well as negative gate voltage
   - iv. none of the above

2. Which power amplifier has the highest collector efficiency
   - i. C
   - ii. A
   - iii. B
   - iv. AB

3. Which amplifier is used for 180-degree phase shift
   - i. differential amplifier
   - ii. non inverting amplifier
   - iii. inverting amplifier
   - iv. None of the above

4. A MOSFET has three terminals, namely
   - i. cathode, anode, grid
   - ii. source, gate, drain
   - iii. emitter, base, collector
   - iv. none of the above

5. The maximum efficiency of resistance loaded class B power amplifier is
   - i. 5%
   - ii. 50%
   - iii. 30%
   - iv. None of the above

6. A JFET is also called _____ transistor
   - i. unipolar
   - ii. bipolar
   - iii. injunction
   - iv. none of the above

7. In a phase shift oscillator, the frequency determining elements are
   - i. L and C
   - ii. R, L and C
   - iii. R and C
   - iv. None of the above

8. The leakage current across a pn junction is due to
   - i. Minority carriers
   - ii. Majority carriers
   - iii. Junction capacitance
   - iv. None of the above

9. An oscillator converts
   - i. a.c. power into d.c. power
   - ii. a.c. power into a.c. power
   - iii. mechanical power into a.c. power
   - iv. none of the above

10. In a transistor
    - i. $I_E = I_C - I_B$
    - ii. $I_B = I_C + I_E$
    - iii. $I_C = I_E + I_B$
    - iv. $I_E = I_C + I_B$

Q.2 Solve both questions : **[14]**

1. In a full wave rectifier, the input is from 30-0-30V transformer. The load and diode forward resistances are 100Ω and 10 Ω respectively. Calculate the average voltage, dc output power, ac input power, rectification efficiency and percentage regulation.

2. Explain the working of positive clamping circuit.

Q.3 Solve both questions : **[14]**

1. Draw the common emitter circuit and sketch the input and output characteristics. Also explain active region, cutoff region and saturation region by indicating them on the characteristic curve.

2. With a neat circuit diagram explain the Voltage Divider Bias circuit by giving its exact analysis.

Q.4 Solve both questions : **[14]**

1. Explain the characteristics of an Ideal Op-Amp. Mention some of the applications of Op-Amp.

2. Explain R-2R ladder DAC with example?

Q.5 Solve both questions : **[14]**

1. Explain Successive type ADC with neat circuit diagram.

2. Describe the construction and operation of JFET and draw its characteristics.

Q.6 Solve both questions : **[14]**

1. Explain Colpitts oscillator with Circuit diagram and derive its frequency.

2. Explain RC coupled transistor amplifier with neat circuit diagram in a middle frequency Range.

Q.7 Solve both questions : **[14]**

1. Explain RC- phase shift oscillator using op-amp and derive its frequency.

2. Describe the construction and operation of depletion type MOSFET and draw its characteristics.

Q.8 Solve both questions : **[14]**

1. Describe class A Power Amplifier. Derive the expression for its maximum conversion Efficiency.

2. Explain Wien bridge oscillator using op-amp and derive its frequency.

Q.9 Solve both questions : **[14]**

1. Explain MOSFET as an amplifier?

2. Describe class B Power Amplifier. Derive the expression for its maximum conversion Efficiency.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023 Semester-IV

ECE ANALOG CIRCUIT

Time: 03 Hours | Code: 104401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Write the answer of the following (Any seven question only): **[14]**

1. Which of these are incorrect about Darlington amplifier?
   i. It has a high input resistance
   ii. The output resistance is low
   iii. It has a unity voltage gain
   iv. It is a current buffer

2. What is a cascade amplifier?
   i. A cascade of two CE amplifiers
   ii. A cascade of two CB amplifiers
   iii. A cascade of CE and CB amplifiers
   iv. A cascade of CB and CC amplifiers

3. What is the output of a class B amplifier for sinusoidal input?
   i. Sinusoidal amplifier
   ii. Half-sinusoidal
   iii. Sinusoidal with higher frequency
   iv. Square wave

4. In a class B amplifier, it is found that DC power is 25W, find the ac power.
   i. 10 W
   ii. 62.5 W
   iii. 25 W
   iv. 50 W

5. In an oscillator if phase of feedback is same as that of oscillation waveform then feedback is called
   i. Positive feedback
   ii. Negative feedback
   iii. Cannot be predicted
   iv. Either positive or negative depending upon frequency

6. For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40$\mu A/V^2$. Find drain to source current in saturation.
   i. 0.10 mA
   ii. 0.05mA
   iii. - 0.05mA
   iv. -50A

7. For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40$\mu A/V^2$. Find drain to source current in saturation.
   i. 1100 Ω
   ii. 11000 Ω
   iii. 900 Ω
   iv. 110 Ω

8. The hybrid π model is used for
   i. Analysis of BJT for low frequency
   ii. Analysis of BJT at high frequency
   iii. Analysis of BJT for feedback
   iv. Analysis of BJT for large signal

9. Which oscillator is characterized by a split capacitor in its tank circuit?
   i. RC phase shift oscillator
   ii. Colpitts oscillator
   iii. Wien bridge oscillator
   iv. none

10. The gate voltage in a JFET at which drain current becomes zero is called
    i. pinch-off voltage.
    ii. saturation voltage
    iii. active voltage.
    iv. cut-off voltage.

Q.2 Solve this question : **[14]**

1. Explain construction and working of Enhancement and Depletion type MOSFET.

Q.3 Solve this question : **[14]**

1. What is power amplifier. Explain Transformer coupled Class-A amplifier.

Q.4 Solve this question : **[14]**

1. Barkhausen criteria play an important role while designing Oscillator justify. Explain Hartley and RC phase shift Oscillator.

Q.5 Solve this question : **[14]**

1. Explain the effect of voltage series and voltage shunt feedback on gain & impedance of amplifier.

Q.6 Solve this question : **[14]**

1. The BJT is connected as a CE amplifier. Determine Z_i, Z_0, A_v, A_i hybrid model.

Q.7 Solve this question : **[14]**

1. Describe current mirror circuit in BJT. What are the condition to design current mirror circuit. Obtain the expression for current transfer ratio.

Q.8 Solve this question : **[14]**

1. Derive the expression for voltage gain and current gain and B.W of multistage amplifier.

Q.9 Solve this question : **[14]**

1. In brief explain the working of voltage series and current series circuit. Explain the same in case of BJT.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Exam., 2022

ANALOG ELECTRONIC CIRCUITS

Time: 03 Hours | Code: 110401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Attempt any seven from the following: [14]

a. What is avalanche multiplication?

b. Compare CE, CC and CB for current gain, voltage gain, input and output impedance.

c. Junction transistor is an active device. Justify.

d. Write the equation of a diode and draw its VI characteristics.

e. What is early effect in bipolar junction transistors?

f. Draw an npn transistor in CB configuration and draw its input characteristics.

g. BJT has higher driving capability as compared to MOSFET. Justify.

h. For an ideal negative feedback configuration, there is a virtual short circuit between the input terminals. Why?

i. Draw a circuit which can be used as limiter or clamper or both.

Q.2 Solve both questions : **[14]**

1. Write brief notes on: (i) Avalanche breakdown (ii) Zener breakdown

2. The saturation currents of the diodes are 1 μA and 2 μA. The breakdown voltages of the diodes are the same and are equal to 100 V: (i) Calculate the current and voltage for each diode if V = 90 V and V = 110 V (ii) Repeat part (i) if each diode is shunted by 10 MΩ resistor.

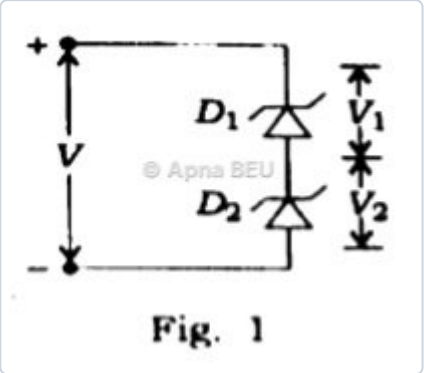


Fig. 1

Q.3 Solve both questions : **[14]**

1. With reference to rectifying circuits, explain- (i) ripple factor (γ); (ii) transformer utility factor (TUF); (iii) peak inverse voltage (PIV); (iv) rectifier efficiency (η).

2. For HW rectifier and capacitor filter- (i) draw the circuit diagram; (ii) draw the waveforms; (iii) give a brief explanation of the functioning of the circuit.

Q.4 Solve both questions : **[14]**

1. Draw Ebers Moll model for pnp transistor and show that transistor action is not possible by connecting two diodes back to back.

2. For the transistor shown below, $\alpha = 0.98$ and $V_{BB} = 0.7$, find R_1 for an emitter current $I_E = -2mA$. Neglect I_{CO}.

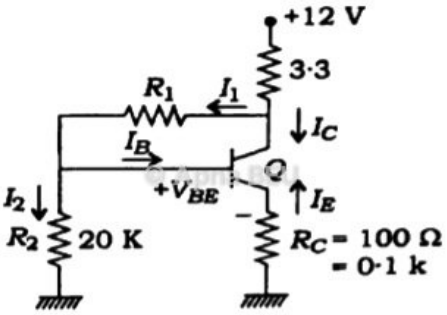


Fig. 2

Q.5 Solve both questions : **[14]**

1. Given the following measurements made at $I_C = 5mA$, $V_{CE} = 10V$ and at room temperature $h_{fe} = 100$, $h_{ie} = 600\Omega$, $A_{ie} = 10$ at 10 MHz, $C_C = 3pF$. Find f_T, f_α, $r_{b'e}$ and $r_{bb'}$ for the transistor.

2. A silicon transistor with $\beta = h_{FE} = 100$, $V_{BE}(sat) = 0.8V$, $V_{CE}(sat) = 0.2V$ is used in the circuit shown. Find the minimum value of R_C for which transistor remains in saturation.

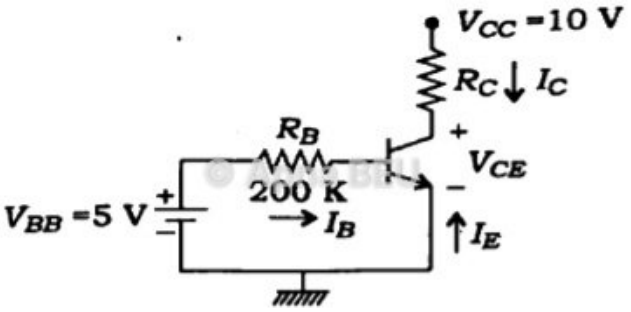


Fig. 3

Q.6 Solve both questions : **[14]**

1. Compare the performance of MOSFET and BJT.

2. Explain the construction of p-channel MOSFET. Draw and explain both depletion and enhancement mode MOSFET.

Q.7 Solve both questions : **[14]**

1. Calculate v_0 and the op-amp's output current as shown in the circuit and if E_i equals (i) +5 V and (ii) -2 V. For each situation, state if the op-amp sources or sinks current.

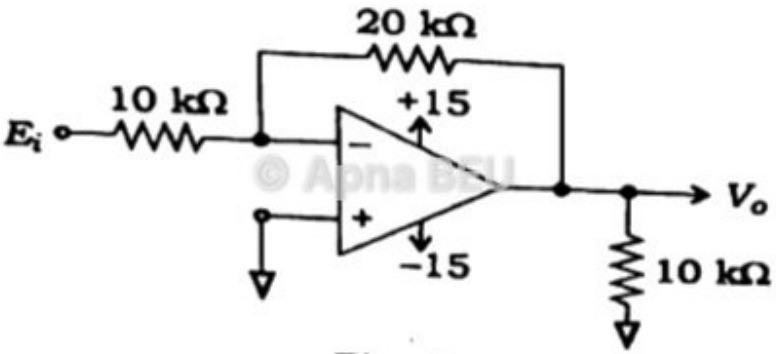
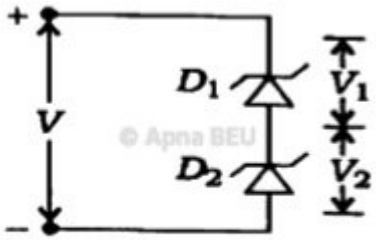


Fig. 4

2. (i) Design an inverting amplifier with a gain of -5 and an input resistance of 10 kΩ. (ii) Design a non-inverting amplifier with a gain of 5.

Q.8 Solve both questions : **[14]**

1. The circuit shown is required to supply on average voltage of 120 V. Determine the r.m.s. value of secondary voltage, average current delivered to the load, peak current through each diode and PIV of diodes.



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Fig. 5

2. Explain different amplifiers with suitable example.

Q.9 Write short notes on the following: **[14]**

a. Precision comparator

b. Zero crossing detector with hysteresis

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 5th Semester Exam., 2020 (New Course)

LINEAR INTEGRATED CIRCUITS AND APPLICATIONS

Time: 03 Hours | Code: 104504 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are EIGHT questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven of the following short answer-type questions : [2×7=14]

a. Write two requirements to be met for a good current source.

b. Why VCO is also called as v-to-f converter?

c. What is a switched capacitor filter?

d. What output voltage would be produced by a D/A converter whose output range is 0 to 10 V and whose input binary number is 0110 for a 4-bit DAC?

e. What is a power amplifier?

f. What is the purpose of connecting a capacitor at the input and the output side of an IC voltage regulator?

g. Mention any two applications of PLL.

h. For an n-bit flash-type A/D converter, how many comparators are required?

i. What is meant by frequency synthesizing?

j. Why are integrators preferred over differentiators?

Q.2 Solve both questions : [7+7=14]

a. With suitable circuit diagram, explain the operation of voltage reference circuit using temperature compensation.

b. What are the different methods used in fabrication of integrated resistors? Explain any one method with suitable diagrams.

Q.3 Solve both questions : [7+7=14]

a. Draw and explain the circuit of a voltage to current converter if the load is (i) floating and (ii) grounded.

b. Define and explain slew rate. What is full-power bandwidth? Also explain the methods to improve the slew rate.

Q.4 Solve both questions : **[7+7=14]**

a. Explain the working of a anti logarithmic amplifier using suitable op-amp circuit diagram and supporting equations.

b. Draw the functional diagram of 723 regulators and explain its working.

Q.5 Solve both questions : **[7+7=14]**

a. Draw the block diagram and explain the principle of working, characteristic and applications of a frequency shift keying (FSK) demodulator.

b. Explain the operation of frequency to voltage converter with suitable circuit diagram.

Q.6 Solve both questions : **[7+7=14]**

a. Explain the working of R-2R ladder D/A converter with circuit diagram.

b. Explain the oversampling A/D converter with functional block diagram.

Q.7 Solve both questions : **[7+7=14]**

a. Explain with block diagram how PLL is used as frequency modulation (FM) detector.

b. Design a VCO having the maximum range of 1 kHz. Assume power supply $V_{cc} = 15\ \mathrm{V}$. Required pulse width should not exceed 100 $\mu\mathrm{s}$.

Q.8 Solve both questions : **[7+7=14]**

a. Design an astable multivibrator circuit which will flash the electric bulb such that its on time will be 4 seconds and off time will be 2 seconds.

b. Explain the working of Schmitt trigger using 555 timer.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

DIGITAL CIRCUITS

Time: 03 Hours | Code: 104403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Write the short answer of the following (Any seven question only): **[14]**

1. Find the hexadecimal equivalent of the decimal number 256.
2. What is meant by weighted and non-weighted coding?
3. State Demorgan's theorem.
4. What do you mean by carry propagation delay?
5. What do you mean by Encoder?
6. What are the primary disadvantages of an asynchronous counter?
7. What are races and cycles?
8. Why is MUX called as data selector?
9. When can RTL be used to represent digital system?
10. Draw the DTL based NAND gate.

Q.2 Solve both questions : **[10+4=14]**

1. Describe the procedure to design of asynchronous sequential circuit in detail with an example.
2. What are the disadvantages of asynchronous sequential circuit?

Q.3 Solve both questions : **[10+4=14]**

1. Explain the characteristics and about TTL parameters. Discuss implementation of the following digital logic families : (a) CMOS (b) TTL
2. Discuss about TTL parameters.

Q.4 Solve both questions : **[7+7=14]**

1. Explain the realization of JK flipflop from T flipflop.
2. Design a 4-bit BCD ripple counters.

Q.5 Solve this question : **[14]**

1. What is meant by magnitude comparator? Discuss in detail about two bit magnitude comparator.
2. Design a full adder using NOR gates.

Q.6 Solve both questions : **[7+7=14]**

1. Explain briefly about SOP and POS forms with example.
2. Plot the logical expression $ABCD + A\overline{B}\overline{C}\overline{D} + A\overline{B}C + AB$ on a 4 variable K-map. Obtain the simplified expression from the map.

Q.7 Solve both questions : **[7+7=14]**

1. How does don't care condition in K-map help for circuit simplification?
2. Write the POS form of the SOP expression $f(x, y, z) = \overline{X}YZ + XY\overline{Z} + X\overline{Y}Z$.

Q.8 Solve both questions : **[7+7=14]**

1. Implement the following function using suitable multiplexer. $F(a, b, c) = \Sigma m(3, 4, 5, 7)$.
2. Implement JK flipflop using SR flipflop.

Q.9 Write Short notes on any two of the following: **[14]**

1. (a) Analog to Digital Converter
2. (b) Multivibrator
3. (c) Monostable and Astable timer

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer of the following questions (any seven only): [14]

1. What is the use of K-map?
2. Discuss Universal gates.
3. Add hexadecimal number 2ABC & 98F2.
4. Number of 2:1 mux requires designing 256:1 mux is .....
5. Find 2's complement of 1011011.
6. What is the function of a sample-and-hold circuit?
7. Define term propagation delay.
8. Define race around condition in JK flip flop.
9. What is the purpose of expanding memory size?
10. Differentiate between ROM and RAM.

Q.2 Solve both questions : [14]

1. Realize XNOR logic function using NAND gate only.
2. Simplify $Y = ABC + AB\overline{C} + A\overline{B}C$

Q.3 Solve this question : [14]

1. A logic circuit has four inputs A, B, C, D and output Y. Y = 1 when A & B are both 1 subjected to the condition that C and D are both low or both high. Design the logic circuit.

Q.4 Solve both questions : **[14]**

1. Design 8 to 3 line Encoder circuit.

2. Implement NAND gate using TTL logic family.

Q.5 Solve both questions : **[14]**

1. Design a parallel-to-serial converter using shift registers and explain its operation.

2. Compare the characteristics and applications of JK and T flip-flops.

Q.6 Solve both questions : **[14]**

1. Explain the working principle of an R-2R ladder DAC with a detailed diagram.

2. Design a 3-bit flash ADC and explain its working with an example.

Q.7 Solve both questions : **[14]**

1. Find the Simplified logical expression for Y.
   $Y(A, B, C, D, E) = \Sigma m$(0, 2, 4, 7, 8, 10, 12, 16, 18, 20, 23, 24, 25, 26, 27, 28)

2. Summarize the design procedure for a synchronous sequential circuit.

Q.8 Solve both questions : **[14]**

1. Implement S-R, T, D flip-flops using J-K flip-flop. Also show the implementation with help of State Tables.

2. Discuss the organization and operation of content-addressable memory (CAM).

Q.9 Write short notes on the following: **[14]**

1. (a) Binary Parallel Adder
   (b) Digital IC logic families

BIHAR ENGINEERING UNIVERSITY, PATNA

Special Examination - 2023

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven question only: [14]

1. Define Dynamic RAM.
2. The resolution of a 10-bit AD converter for an input range of 10 V is approximately _____
3. The parameter through which 16 distinct values can be represented is known as _____
4. Define Multiplexing.
5. Write the significance of Truth table.
6. Write the differences between Combinational and Sequential circuit.
7. A hexadecimal odometer displays F52F. The next reading will be _____
8. The basic property of Sequential circuit is _____
9. How many entries will be in the truth table of a 4-input NAND gate?
10. What is Dual Slope ADC?

Q.2 Solve this question : [14]

1. What are weighted, non-weighted, cyclic and self-complementary codes? Explain each with examples.

Q.3 Solve this question : [14]

1. Design a BCD to 7-segment display decoder circuit using logic gates.

Q.4 Solve this question : [14]

1. Design a 3-bit parallel comparator A/D convertor that provides output in 2's complement format.

Q.5 Solve both questions : **[14]**

1. Design full adder using 8:1 mux.

2. Design full adder using 4:1 mux.

Q.6 Solve this question : **[14]**

1. What is binary shift register? Write down their application.

Q.7 Solve this question : **[14]**

1. Minimize the following expression using k-map
   $f(P, Q, R, S) = \Sigma m(0, 1, 4, 5, 7, 12, 13)$

Q.8 Solve this question : **[14]**

1. Describe the procedure to design Mod-6 counter.

Q.9 Write short notes on any two of the following: **[14]**

1. i. RAM
   ii. ROM
   iii. CMOS logic
   iv. Operation of TTL logic circuit working as NAND Gate

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven Question only: [14]

1. Draw Truth table of JK flip flop.
2. Convert the Decimal number (108.025) to their binary equivalent.
3. Draw the truth table and logic circuit of Half-adder.
4. Write the difference between combinational & Sequential circuit.
5. Draw timing diagram of SR flip-flop.
6. Find 2's complement of 1011011.
7. Number of 2:1 mux requires designing 256:1 mux is
8. Add hexadecimal number 2ABC & 98F2.
9. Discuss Universal gates.
10. What is the use of K-map?

Q.2 Solve this question : [14]

1. Implement the following function using only NAND gate $F(A,B,C) = \Sigma m(0,1,2,3,7)$.

Q.3 Solve both questions : [7+7=14]

1. Realize XNOR logic function using NAND gate only.
2. Simplify $Y = ABC + AB\overline{C} + A\overline{B}C$

Q.4 Solve this question : [14]

1. A logic circuit has four inputs A,B,C,D and output Y. Y=1 when A & B are both 1, subjected to the condition that C and D are both low or both high. Design the logic circuit.

Q.5 Solve this question : **[14]**

1. Explain Master-slave flip flop. What are race around condition?. How it can be circumvented with the help of Master-slave flip flop.

Q.6 Solve both questions : **[7+7=14]**

1. Design 8 to 3 line Encoder circuit.

2. Implement NAND gate using TTL logic family.

Q.7 Solve this question : **[14]**

1. Summarize the design procedure for a synchronous sequential circuit.

Q.8 Solve both questions : **[7+7=14]**

1. Explain the working of R-2R Ladder DAC.

2. Design 3-bit binary counter using T flip-flop.

Q.9 Write short notes on the following: **[14]**

1. (a) Binary Parallel Adder

2. (b) Digital IC logic families

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

ELECTROMAGNETIC THEORY

Time: 03 Hours | **Code: 104404** | **Full Marks: 70**

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[14]**

1. The divergence of the vector field is
   i. Vector
   ii. Scalar
   iii. Can't be both
   iv. None

2. The unit of electric field intensity
   i. Volt
   ii. Volt per metre
   iii. Tesla
   iv. Metre

3. In Gauss's Law the electric field is related to
   i. The charge density
   ii. The potential difference
   iii. The charge enclosed
   iv. The capacitance

4. The electric field at a point due to an infinite sheet of charge is:
   i. $\frac{\sigma}{\epsilon_0}$
   ii. $\frac{\sigma}{2\epsilon_0}$
   iii. $\frac{2\epsilon_0}{\sigma}$
   iv. 0

5. The force on a charged particle moving in a magnetic field is maximum when the angle between the velocity and magnetic field is:
   i. $0°$
   ii. $45°$
   iii. $90°$
   iv. $180°$

6. The energy density in magnetic field is given by
   i. $\frac{\mu H^2}{2}$
   ii. $\frac{H^2}{2\mu}$
   iii. 2
   iv. None

7. The dot product of the vectors $3i - 2j + 5k$ and $-i + 3j + 2k$ is,
   i. -1
   ii. 1
   iii. 0
   iv. -3i

8. The Pointing vector P is
   i. $E \times H$
   ii. $E \cdot H$
   iii. $\frac{1}{2}E \times H$
   iv. $(E \times H)^2$

9. The Maxwell's equation $\nabla \cdot B = 0$ is due to
   i. Monopole does not exit
   ii. Coulomb's Law
   iii. Existence of monopoles
   iv. Ampere Circuital Law

10. Curl of gradient of a vector is
    i. 0
    ii. Null Vector
    iii. Unity
    iv. Depends on the constants of the vector

Q.2 Solve both questions : **[7+7=14]**

1. Write the differential elements (dl, da, dv) in both Cartesian, cylindrical co-ordinate system.
2. State divergence theorem. What will be divergence to position vector?

Q.3 Solve both questions : **[7+7=14]**

1. Given the two points, $C(-3,2,1)$ and $D(r=5,\theta=20°,\varphi=-70°)$. find: (i) The spherical coordinates of C (ii) The Cartesian coordinates of D
2. Find the divergence of $\vec{A}=3x^2a_x+5x^2y^2a_y+xyz^3a_z$ where a_x, a_y and a_z are unit vectors in cartesian coordinates at point (1,1,1)

Q.4 Solve both questions : **[7+7=14]**

1. State and derive Coulomb's law. Write coulomb's law in vector forms.
2. Derive an expression for intensity of electric field at a point distant r from a point charge.

Q.5 Solve both questions : **[7+7=14]**

1. An infinite long line charge of uniform density ρ_L C/cm is situated along the z-axis. Obtain electric field intensity due to this charge using Gauss's law.
2. Derive energy density in electrostatic field.

Q.6 Solve both questions : **[7+7=14]**

1. Four 3pC charges are at the corners of a 1-m square. The two charges at the left side of the square are positive. The two charges on the right side are negative. Find the field E at the centre of the square, $\epsilon_r=1$
2. What do you understand by capacitance of a capacitor? Deduce and expression for the capacitance of a parallel plate capacitor. How will it be modified when the gaps between the plates is filled with a dielectric?

Q.7 Solve both questions : **[7+7=14]**

1. Show that the Faraday's law of electromagnetic induction can be expressed as $\nabla\times E=-\partial B/\partial t$. Write down its integral form.
2. Explain the concept of displacement current and show how it led to the modification of the Ampere's law.

Q.8 Solve both questions : **[7+7=14]**

1. Discuss reflection of plane electromagnetic wave incident normally on a perfect dielectric and obtain expressions for the two reflection coefficients of electric and magnetic fields.
2. Define Poynting vector. Mention any two properties of uniform plane wave.

Q.9 Write short notes on any two of the following:- **[14]**

a. Biot-Savart's Law

b. Magnetic field boundary condition for isotropic medium

c. Skin Effect

d. Spherical capacitance

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

ELECTROMAGNETIC THEORY

Time: 03 Hours | Code: 104404 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven question only: [14]

a. What is physical significance of divergence?

b. State Stoke's theorem.

c. How is the unit vectors defined in three co ordinate systems?

d. State Gauss law for electric fields.

e. Define dielectric strength.

f. State Biot-Savarts law.

g. What is magnetic susceptibility?

h. Write down the wave equation for E and H in free space.

i. State Poynting Theorem.

j. Define intrinsic impedance or characteristic impedance.

Q.2 Solve both questions : [7+7=14]

1. Derive the expression for the attenuation constant, phase constant and intrinsic impedance for a uniform plane wave in a good conductor.

2. Calculate the depth of penetration in copper at 10 MHZ. Given the conductivity of copper is $5.8 \times 10^7 S/m$ and its permeability $1.3 mH/m$.

Q.3 Solve both questions : [7+7=14]

1. Derive suitable relations for integral and point forms of poynting theorem.

2. Discuss about the plane waves in lossless dielectrics.

Q.4 Solve this question : **[14]**

1. With explanation, derive the Maxwell's equation in differential and integral forms.

Q.5 Solve both questions : **[7+7=14]**

1. An iron ring with a cross sectional area of 3 cm^2 and mean circumference of 15 cm is wound with 250 turns wire carrying a current of 0.3A. The relative permeability of ring is 1500. Calculate the flux established in the ring.
2. Derive the expressions for boundary conditions in magnetic fields.

Q.6 Solve both questions : **[7+7=14]**

1. State and proof gauss law and explain applications of Gauss law.
2. Explain Poisson's and Lapace's equations.

Q.7 Solve both questions : **[7+7=14]**

1. State and proof divergence theorem.
2. Define divergence, gradient, curl in spherical co-ordinate system with mathematical expression

Q.8 Solve both questions : **[7+7=14]**

1. Define and explain Biot-Savart Law.
2. Derive General field relation for time varying electric and magnetic fields using Maxwell's equations.

Q.9 Write short notes on any two of the following:- **[14]**

a. Equipotential Surface.

b. Magnetic Vector potential.

c. UHF Transmission line.

d. Smith Chart.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 3rd Semester Exam., 2020 (New Course)

ELECTROMAGNETIC FIELD THEORY

Time: 3 hours | Code: 100308 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.
5. Symbols and notations carry their usual meanings.

Q.1 Choose the correct answer of any seven of the following: **[14]**

1. In free space, the Poisson equation becomes
   - i. Maxwell equation
   - ii. Ampere equation
   - iii. Laplace equation
   - iv. steady-state equation

2. Poisson equation can be derived from which of the following equations?
   - i. Point form of Gauss law
   - ii. Integral form of Gauss law
   - iii. Point form of Ampere law
   - iv. Integral form of Ampere law

3. Poynting vector gives the
   - i. rate of energy flow
   - ii. direction of polarization
   - iii. intensity of electric field
   - iv. intensity of magnetic field

4. Using volume integral, which quantity can be calculated?
   - i. Area of cube
   - ii. Area of cuboid
   - iii. Volume of cube
   - iv. Distance of vector

5. Electric flux density in electric field is referred to as
   - i. number of flux lines
   - ii. ratio of flux lines crossing a surface and the surface area
   - iii. direction of flux at a point
   - iv. flux lines per unit area

6. Which of the following correctly states Gauss law?
   - i. Electric flux is equal to charge
   - ii. Electric flux per unit volume is equal to charge
   - iii. Electric field is equal to charge density
   - iv. Electric flux per unit volume is equal to volume charge density

7. Find the power reflected in a transmission line, when the reflection coefficient and input power are 0.45 and 18 W respectively.
   - i. 3.645
   - ii. 6.453
   - iii. 4.563
   - iv. 5.463

8. In a waveguide, which of the following conditions is true always?
   - i. Phase velocity = c
   - ii. Group velocity = c
   - iii. Phase velocity > c
   - iv. Phase velocity < c

9. The phase and group velocities do not depend on which of the following?
   - i. Frequency
   - ii. Wavelength
   - iii. Phase constant
   - iv. Attenuation constant

Q.2 Attempt any two parts of the following: **[7x2=14]**

1. A rectangular waveguide with dimensions of $3\ \text{cm} \times 2\ \text{cm}$ operates at 10 GHz. Find: (i) cut-off frequency (f_c); (ii) cut-off wavelength (λ_c); (iii) guided wavelength (λ_g); (iv) phase constant (β_g).
2. What do you mean by transmission line? Derive an expression for transmission line equations.
3. Determine the expression for average power of Poynting vector.

Q.3 Attempt any two parts of the following: **[7x2=14]**

1. (i) Define quality factor. Give its relation with attenuation factor. (ii) Define reflection coefficient and VSWR. Also write their interrelation.
2. (i) Compare wave impedance and characteristic impedance. (ii) Define tangent loss.
3. Derive the field components when wave is propagating inside a rectangular waveguide with TM mode propagation.

Q.4 Attempt any two parts of the following: **[7x2=14]**

1. Derive an expression for input impedance when transmission line is terminated with any load impedance.
2. What is equipotential surface? Explain Poynting vector and average Poynting vector.
3. State and prove Ampere's work law as $\nabla \times \vec{H} = J$.

Q.5 Attempt any two parts of the following: **[7x2=14]**

1. Derive the Gauss divergence theorem and Stokes' theorem along with their significances.
2. Explain the wave between parallel planes. Derive the expression for the attenuation in parallel plane guide.
3. Derive the expressions for the reflection and refraction of the waves by the perfect dielectric.

Q.6 Attempt any two parts of the following: **[7x2=14]**

1. Find the reflection and transmission coefficient for the interface between air and freshwater $\epsilon + j180$ in the case of perpendicular incidence.
2. Derive the relationship between the following: (i) Standing-wave ratio and magnitude of reflection coefficient. (ii) Standing-wave ratio and the reflection coefficient.
3. (i) Write the condition for a line to be distortionless. (ii) Define the term 'phase velocity'.

Q.7 Attempt any two parts of the following: **[7x2=14]**

1. What is polarization of wave? Discuss the properties of S- and P-polarized light. Explain why P-polarized light is also called as TM-polarized light.

2. Explain the term 'standing-wave ratio' related to transmission line. What will be the values of input impedances when output impedances are (i) open-circuited and (ii) short-circuited?

3. Explain why TEM wave does not propagate in waveguide.

Q.8 Attempt any two parts of the following: **[7x2=14]**

1. A transmission line has a characteristic impedance of 100 ohms and is terminated in a load impedance of $200 + j180$ ohms. Find the voltage reflection coefficient.

2. What is the penetration depth in current penetration in copper at a frequency of 10^4 MHz, if the resistivity is $1.7 \times 10^{-6}\Omega$ cm?

3. What are the satisfactory conditions for low-loss transmission lines?

Q.9 Attempt any two parts of the following: **[7x2=14]**

1. A uniform plane wave propagating in a medium has $E = 2e^{-\alpha z}\sin(10^8 t - \beta z)a_y$. If the medium is characterized by $\epsilon_r = 1$, $\mu_r = 20$ and $\sigma = 3\ \text{mhos/m}$, then find α, β and $\vec{H}$.

2. In a non-magnetic medium $E = 4\sin(2\pi \times 10^7 - 0.8x)a_z\ \text{V/m}$. Find (i) the time-average power carried by the wave; (ii) the total power crossing $100\ \text{cm}^2$ of plane $2x + y = 5$.

3. What is the boundary condition for metal dielectric interface?

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

SEMICONDUCTOR PHYSICS AND DEVICES

Time: 03 Hours | Code: 104405 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[2x7=14]**

a. Which of the following best describes the energy band model of a semiconductor?

i. Electrons are localized within individual atoms
ii. Electrons occupy energy bands separated by a band gap
iii. Electrons are free to move without any restriction
iv. All energy levels are completely filled

b. In an extrinsic semiconductor at equilibrium, the Fermi level E_F shifts towards:

i. The conduction band in p-type material
ii. The conduction band in n-type material
iii. The valence band in n-type material
iv. The mid-gap energy level in doped semiconductors

c. Which of the following phenomena describe the generation and recombination processes in semiconductors?

i. Band-to-band recombination
ii. Impact ionization
iii. Auger recombination
iv. All of the above

d. What is the primary cause of the built-in potential in a PN junction diode?

i. Applied external voltage across the junction
ii. Diffusion of majority carriers
iii. Recombination of electrons and holes
iv. Ionization of donor and acceptor atoms

e. Which of the following mechanisms is responsible for junction breakdown in a Zener diode?

i. Avalanche multiplication
ii. Thermal runaway
iii. Tunneling of carriers across the junction
iv. Both (i) and (iii)

f. In which mode of operation does a Bipolar Junction Transistor (BJT) act as an amplifier?

i. Cutoff mode
ii. Active mode
iii. Saturation mode
iv. Reverse-active mode

g. What is the expression for common-base current gain (α) of a BJT?

i. $\alpha = I_C/I_B$
ii. $\alpha = I_E/I_C$
iii. $\alpha = I_C/I_E$
iv. $\alpha = I_B/I_C$

h. In a Junction Field-Effect Transistor (JFET), the drain current (I_D) in the saturation region is primarily controlled by:

i. The gate-to-source voltage (V_{GS})
ii. The source current (I_S)
iii. The drain-to-source voltage (V_{DS})
iv. The drain resistance

i. Which of the following correctly describes the threshold voltage (V_{th}) of a MOSFET?

i. The voltage at which the drain current becomes zero
ii. The voltage at which the device enters saturation
iii. The minimum gate voltage required to form an inversion layer at the semiconductor-oxide interface
iv. The voltage at which the oxide layer begins conducting current

j. Which of the following devices is primarily used for triggering other semiconductor devices like SCRs?

i. TRIAC
ii. UJT
iii. DIAC
iv. MOSFET

Q.2 Solve both questions : **[14]**

a. Explain the concept of the bonding model and energy band model in semiconductors.

b. Explain drift and diffusion current mechanisms in semiconductors. Derive the Einstein relation that connects diffusion coefficient and mobility.

Q.3 Solve both questions : **[14]**

a. Discuss the different types of recombination mechanisms and their significance in semiconductor devices.

b. Write and explain the continuity equation for charge carriers in semiconductors. Derive the minority carrier diffusion equation and discuss its importance in the operation of semiconductor devices.

Q.4 Solve both questions : **[14]**

a. Explain the Ideal Diode Equation. Derive it quantitatively using the band model and boundary conditions, stating the assumptions made in the derivation.

b. Explain the working principles and applications of special diodes such as the Zener diode, Tunnel diode, Varactor diode and Schottky diode. How do their characteristics differ from a conventional PN junction diode?

Q.5 Solve this question : **[14]**

Discuss the two primary mechanisms of junction breakdown in a PN junction diode- Avalanche breakdown and Zener breakdown. Compare their occurrence conditions and characteristics.

Q.6 Solve this question : **[14]**

Describe the impact of base width modulation (Early effect) on BJT operation. How does it affect the common-emitter current gain and output characteristics? Include relevant equations in your explanation.

Q.7 Solve this question : **[14]**

Explain the deviations from ideal MOS behaviour, including metal-semiconductor work function differences, fixed oxide charges, interface traps and threshold voltage shifts. How do these non-ideal effects influence MOSFET performance?

Q.8 Solve this question : **[14]**

Discuss the working principles and characteristics of different optoelectronic devices: PIN Photodiode, Avalanche Photodiode, Solar Cell and LED. Compare their applications.

Q.9 Solve this question : **[14]**

Explain the working of a solid-state laser diode. How does it differ from an LED in terms of construction, working principle, and applications? Discuss its advantages in optical communication.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023 (Semester IV)

SEMICONDUCTOR PHYSICS AND DEVICES

Time: 03 Hours | Code: 104405 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Write the short answer of the following (Any seven question only): **[2x7=14]**

a. Define the term drift velocity.

b. Define Fermi energy level.

c. Define doping process.

d. Why is the width of the base region kept very small as compared to other regions in BJT's?

e. Write the difference between Zener and Avalanche breakdown.

f. What do you mean by solid state?

g. Define the effective mass and express it in terms (E, K).

h. Define conductivity and mobility.

i. Write two differences between BJT and FET.

j. Define the varactor diode.

Q.2 Solve both questions : **[14]**

a. Explain (i) Solar Cell (ii) LED (iii) LASER diode.

b. Discuss the working principle of SCR. How it is different from UJT?

Q.3 Solve both questions : **[14]**

a. What is MOS Capacitor? Draw its C-V Characteristic.

b. Explain the operation of enhancement mode MOSFET with neat diagram and compare it with JFET.

Q.4 Solve both questions : **[14]**

a. Discuss the punch-through breakdown in BJT. Write the expression for the punch though voltage.

b. Distinguish between the different types of transistor configurations with necessary circuit diagrams.

Q.5 Solve both questions : **[14]**

a. With a neat diagrams explain the working of a PN junction diode in forward bias and reverse bias and plot the V-I characteristics.

b. Discuss about the effects of temperature on PN junction diode's Forward and Reverse characteristics.

Q.6 Solve both questions : **[14]**

a. What is Zener diode. Explain V-I characteristics of Zener diode.

b. When a transistor is used as a switch, in which region of output characteristics it is operated? Discuss with reason.

Q.7 Solve this question : **[14]**

What is intrinsic semiconductor and explain the formation extrinsic semiconductors through doping?

Q.8 Solve both questions : **[14]**

a. Derive the expression for current generated due to drifting of charge carriers in semiconductors in the presence of electric field.

b. Derive Einstein's relation in semiconductors?

Q.9 Write Short notes ver any two of the following: **[7x2=14]**

a. charge carriers generation and recombination

b. UJT relaxation Oscillator

c. Schottky diode

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 6th Semester Examination, 2024

ELECTRONICS INSTRUMENTS AND MEASUREMENT

Time: 03 Hours | Code: 104604 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

1. The high torque to weight ratio in an analog indicating instrument indicates
   - i. High Friction loss
   - ii. Low friction loss
   - iii. Nothing as regards friction
   - iv. None of the above

2. Which of the following is the most popular method for measuring low resistance?
   - i. ducter ohmmeter method
   - ii. kelvin double bridge method
   - iii. ammeter-voltmeter method
   - iv. potentiometer method

3. Which of the following method is used for the measurement of Medium Resistance?
   - i. Direct-Deflection method
   - ii. Anderson Bridge
   - iii. Carey-Foster bridge method
   - iv. Kelvin's double bridge method

4. Which of the following is a balance equation for computing the resistance?
   - i. $R_1 = R_2/R_4$
   - ii. $R_1 = R_3/R_4$
   - iii. $R_1 = R_2R_3$
   - iv. $R_1 = R_2R_3/R_4$

5. A Schering bridge can be used for
   - i. measuring voltages
   - ii. testing capacitors
   - iii. protecting the circuit from temperature rises
   - iv. measuring currents

6. What is the dependence of frequency on the balance equation?
   - i. varies by a factor of 2
   - ii. depends on the detector used
   - iii. independent
   - iv. depends on the supply magnitude

7. Which of the following can be measured using Maxwell's Inductance Bridge?
   - i. capacitance
   - ii. frequency
   - iii. mutual Induction
   - iv. Induction

8. The windings of a C.T. are
   - i. tied together
   - ii. Shorted
   - iii. wound over one another
   - iv. Grounded

9. Which of the following device is used to measure power in A.C. circuits?
   - i. ammeter
   - ii. Wattmeter
   - iii. voltmeter
   - iv. ohmmeter

10. When the moving coil in a Dynamometer type wattmeter deflects
    - i. pointer doesn't move
    - ii. current flows
    - iii. voltage is generated
    - iv. pointer moves

Q.2 Solve both questions : **[14]**

1. Explain how the unknown frequency is measured using Wien bridge method?

2. What is the need for standards of measurements? How are they classified?

Q.3 Solve both questions : **[14]**

1. How do you measure the unknown inductance using Hay's Bridge?

2. State the purpose of calibration of measuring instruments. Explain the procedure of calibration of D.C. Voltmeter by using D.C. Potentiometer.

Q.4 Solve both questions : **[14]**

1. How do you measure the unknown capacitance using Schering Bridge? Also draw the phasor diagram.

2. Draw the possible methods of connecting the pressure coil of a wattmeter and compare the errors. Explain the meaning of 'compensating winding' in a wattmeter and show how they help to reduce the error.

Q.5 Solve both questions : **[14]**

1. List out comparisons between CT's and PT's (Any four).

2. A current transformer with a bar primary has 300 turns in its secondary. Resistance and reactance of the secondary circuit are 1.5 Ω and 1.0 Ω respectively. Including the transformer winding with 5 A flowing in the secondary winding, find the phase angle error. (Assuming magnetizing mmf to be 100 AT and core loss to be 1.5 W).

Q.6 Solve both questions : **[14]**

1. Describe the construction and principle of operation of D' Arsonval type Galvanometer.

2. Write short notes on (a) Kelvin's double bridge (b) Digital frequency meter.

Q.7 Solve both questions : **[14]**

1. Describe the measurement of frequency, phase angle and time delay using oscilloscope with suitable diagrams and mathematical expressions.

2. A moving coil instrument gives a full scale deflection of 10mA, when the potential difference across its terminal is 100mV. Calculate (1) The shunt resistance for a full scale deflection corresponding to 100A (2) The resistance for full scale reading with 1000V. Calculate the power dissipation in each case?

Q.8 Solve both questions : **[14]**

1. Draw the block diagram of an electronic voltmeter and explain its operation.

2. Explain BH curve with a neat diagram. What is Hysteresis loop?

Q.9 Solve both questions : **[14]**

1. Prove that for electrodynamometer type wattmeter true power = $\frac{\cos\Phi}{\cos\Phi\cos(\Phi-\beta)} \times$ **actual wattmeter reading**.
   Where $\cos\phi$ = power factor of the circuit, $\beta = \tan^{-1}(wL/R)$ where L and R are the inductance and resistance of the pressure coil.

2. Explain with a neat sketch, the construction and working principle of Megger.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 6th Semester Examination, 2024

MEASUREMENTS AND INSTRUMENTATION

Time: 03 Hours | **Code: 110601** | **Full Marks: 70**

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

1. Which system analyzes brain wave activity?
   i. EOG
   ii. EMG
   iii. ECG
   iv. EEG

2. Which type of sensor is used to measure flow?
   i. Potentiometer
   ii. Venturi tube
   iii. Hall effect sensor
   iv. Thermocouple

3. What does "Six Sigma" focus on?
   i. Enhancing resolution
   ii. Improving precision
   iii. Reducing defects in processes
   iv. Minimizing systematic errors

4. What is the mean of the following data set: 4, 6, 8, 10?
   i. 6
   ii. 2
   iii. 8
   iv. 9

5. Which instrument is used for ultrasonic diagnostic measurement?
   i. Thermocouple
   ii. Oscilloscope
   iii. Ultrasound transducer
   iv. Megger

6. What is an example of a non-invasive Cardiac output measurement diagnosis?
   i. Temperature measurement
   ii. Blood pressure measurement
   iii. ECG
   iv. All of the above

7. What is the function of a transducer in bio-medical instruments?
   i. Convert electrical signals into light
   ii. Measure electric current
   iii. Measure pressure
   iv. Convert biological signals into electrical signals

8. What does a Megger measure?
   i. Current
   ii. Insulation resistance
   iii. Capacitance
   iv. Voltage

9. What does a digital multi-meter measure?
   i. Only current
   ii. Only resistance
   iii. Only voltage
   iv. Voltage, current, and resistance

10. What type of instrument is used to measure inductance?
    i. Ohmmeter
    ii. LCR meter
    iii. Ammeter
    iv. Voltmeter

Q.2 Solve both questions: **[14]**

1. Discuss the use of instrument transformers for current and voltage measurement in power systems.
2. Explain the working principle of Maxwell's bridge with phasor diagram.

Q.3 Solve both questions: **[14]**

1. Explain the Hall effect and its application in Hall sensors for current and voltage measurements.
2. Describe the working principle of a Digital Storage Oscilloscope (DSO).

Q.4 Solve both questions: **[14]**

1. Define and differentiate between true value, accuracy, and precision in measurement systems.
2. Explain the concept of True RMS measurement. Why is it important for accurate AC voltage and current measurements?

Q.5 Solve both questions: **[14]**

1. Explain the Six-Sigma approach to quality control in industrial processes.
2. Describe the working principles of temperature transducers thermocouples and RTD.

Q.6 Solve both questions: **[14]**

1. Explain hysteresis and dead-band in measurement systems. How do they influence the performance of transducers?
2. Explain how bio-electric signals like ECG, EEG, and EMG are generated and measured.

Q.7 Solve both questions: **[14]**

1. Discuss the types of errors that occur in measurements. Explain the difference between systematic and random errors with practical examples.
2. Discuss non-invasive diagnostic measurements used in healthcare.

Q.8 Solve both questions: **[14]**

1. Describe the role of sensitivity in measurement systems. How is sensitivity calculated, and what are the consequences of low or high sensitivity in sensors?
2. Explain the use of Meggers in insulation resistance testing. Why is insulation testing crucial in high-voltage systems?

Q.9 Write short notes on any two of the following: **[14]**

1. a. CAT scan techniques
   b. Cardiovascular measurements
   c. Capacitance Measurement with phasor
   d. LVDT

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Exam., 2022

INSTRUMENTATION AND CONTROL

Time: 3 hours | Code: 102404 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[2x7=14]**

1. Which of the following causes a controlled variable to change?
   i. Disturbance
   ii. Change in set point
   iii. Variation in load demand
   iv. All of the above

2. What is the resolution of a DAC with a maximum voltage of 15 volts and five input lines?
   i. 0.4687
   ii. 0.4838
   iii. 0.2343
   iv. 0.2419

3. The derivative function gives the actuator _____.
   i. boost action
   ii. braking action
   iii. Both (i) and (ii)
   iv. None of the above

4. A square wave that is 20 volts at its On state and 0 volt at its Off state will produce an average DC voltage of _____ when duty cycle is 75%.
   i. 15 V
   ii. 10 V
   iii. 7.5 V
   iv. 5 V

5. The resistance of an RTD is:
   i. directly proportional to the temperature
   ii. inversely proportional to the temperature
   iii. directly proportional to the square of temperature
   iv. inversely proportional to the square of temperature

6. Which of the following valves has the greatest flow capacity of all control valves?
   i. Single-seated
   ii. Three-way
   iii. Butterfly
   iv. Ball

7. A unity feedback system has open-loop transfer function as $G(s) = \frac{16}{s(s+16)}$. The natural frequency of the system is:
   i. 16
   ii. 8
   iii. 2
   iv. 4

8. Which of the following represents the application of inductive transducers?
   i. Displacement measurement
   ii. Thickness measurement
   iii. Both displacement and thickness measurements
   iv. None of the above

9. Which of the following can act as a comparator?
   i. Op-amp with negative feedback
   ii. Op-amp with positive feedback
   iii. Op-amp without feedback
   iv. None of the above

10. Transfer function represents the input-output relationship of a system, with initial conditions as:
   i. zero
   ii. positive
   iii. negative
   iv. All of the above

Q.2 Solve both questions : **[14]**

1. Define the following: (i) Transducer (ii) Resolution (iii) Linearity (iv) Accuracy (v) Repeatability (vi) Calibration (vii) Backlash

2. Discuss in detail about the classification and elimination of error.

Q.3 Solve both questions : **[14]**

1. Differentiate among piezoelectric, inductive and capacitance-type transducers.

2. What are pyrometers? Explain with the help of neat sketch.

Q.4 Solve both questions : **[14]**

1. Explain the working of electrical resistance strain gauge. Explain how the resistance change of a strain gauge is measured with the help of Wheatstone bridge. Also, explain the methods of lead wire compensation and temperature compensation in case of strain gauge.

2. Explain the five basic laws of thermocouple behaviour and how these laws are useful in designing thermocouple for temperature measurement.

Q.5 Solve all questions : **[14]**

1. Consider a system with controller G_c, reference input R, and disturbance input D acting on the plant $\frac{1}{s(s+2)}$. If G_c is a proportional controller, $G_c = K_p$, what value of K_p will cause the steady-state error for a unit ramp reference input to be 0.1?

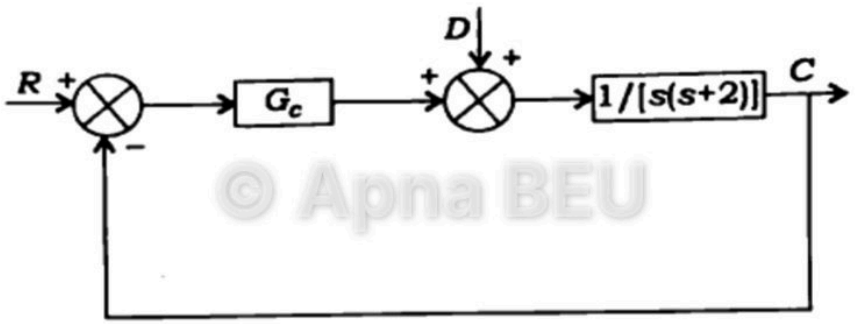
   (a) If G_c is a proportional controller, $G_c = K_p$, what value of K_p will cause the steady-state error for a unit ramp reference input to be 0.1?

   (b) If zero steady-state error for a unit ramp reference input is required, what form should the controller G_c take?

   (c) For a unit-step disturbance input, what form of controller G_c should be chosen to make the steady-state error zero?

   (d) Assume that the reference input is a unit step, the disturbance input is zero and proportional controller being used. What value (or values) of K_p will yield an underdamped system with a settling time of 4 seconds?

   (e) Considering the same case as in part (d), what effect does increasing K_p have on the damping ratio?



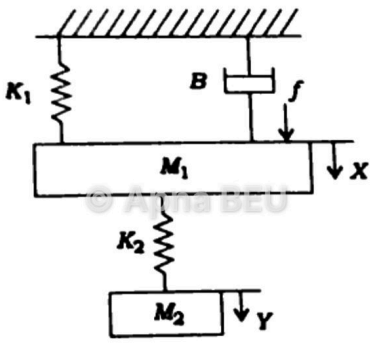
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Figure 1

Q.6 Solve both questions : **[14]**

1. With the help of a suitable example, explain the difference between open-loop control and closed-loop control.

2. For a given mechanical translational system with masses M_1, M_2, springs K_1, K_2, and damper B, find out the transfer function $\frac{X(s)}{F(s)}$.



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Figure 2

Q.7 Solve both questions : **[14]**

1. Explain the operations of the following: (i) Inverting op-amp (ii) Summing op-amp (iii) Differentiating op-amp (iv) Integrating op-amp.

2. What are the important requirements of a signal conditioner and how are they met?

Q.8 Solve this question : **[14]**

1. For the open-loop transfer function $G(s) = \frac{400(s+1)(s+10)}{s^3(s+100)(s+1000)}$, sketch the Bode magnitude and phase diagrams. Sketch the diagrams for each term and then also sketch the total diagrams. Show the slopes of each section and of the total diagram.

Q.9 Write short notes on any four of the following: **[14]**

1. a. Sensitivity analysis
   b. Pneumatic actuators
   c. Flow measurement sensors
   d. ADC using successive-approximation register (SAR)
   e. Optical encoders